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Karlgaard

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(54) **PEAK CONSUMPTION MANAGEMENT FOR RESOURCE DISTRIBUTION SYSTEM**

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(52) **U.S. Cl.**
CPC ***G06Q 10/0631*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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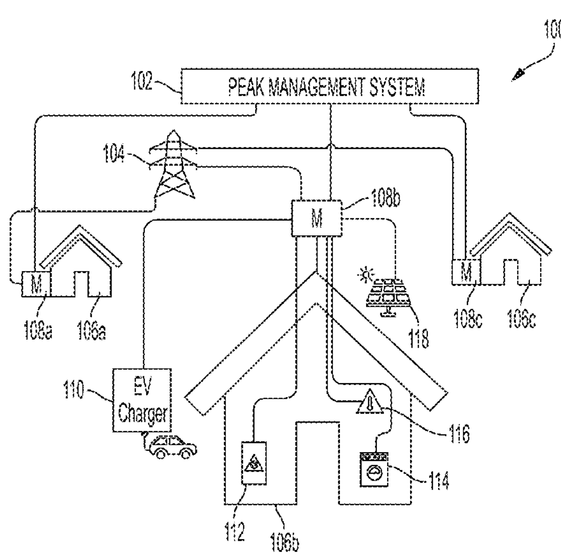
(57) **ABSTRACT**

A computer-implemented method includes assigning, by a peak management system, a peak resource management schedule to a first premises and a second premises within a resource distribution network. The peak resource management schedule includes a first peak consumption period and a first limited resource consumption period for the first premises and a second peak consumption period and a second limited resource consumption period for the second premises, where the first peak consumption period and the second peak consumption period are different. The method also includes transmitting, by the peak management system, a peak resource management signal to the first premises and the second premises. The peak resource management signal includes instructions to commence execution of the peak resource management schedule at the first premises and the second premises.

13 Claims, 8 Drawing Sheets



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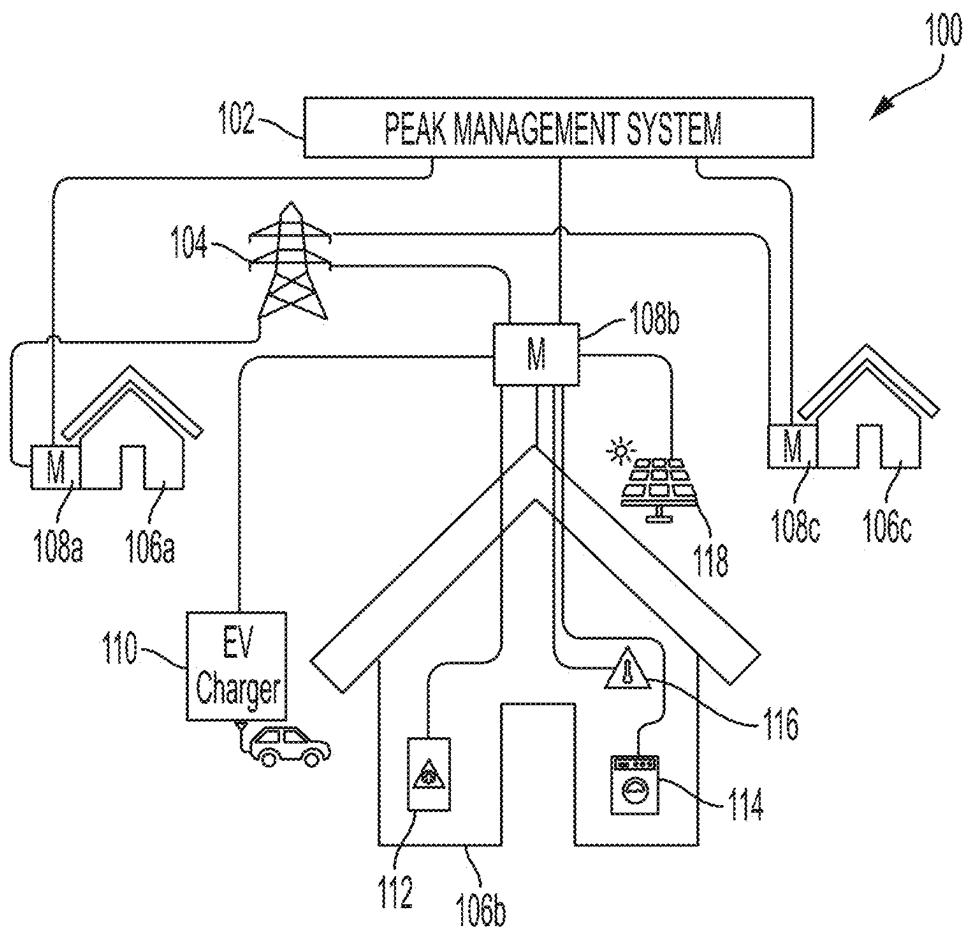
100
102
PEAK MANAGEMENT SYSTEM
104
108b
M
M
M
108a
106a
118
108c
106c
110
EV Charger
116
114
112
106b

FIG. 1

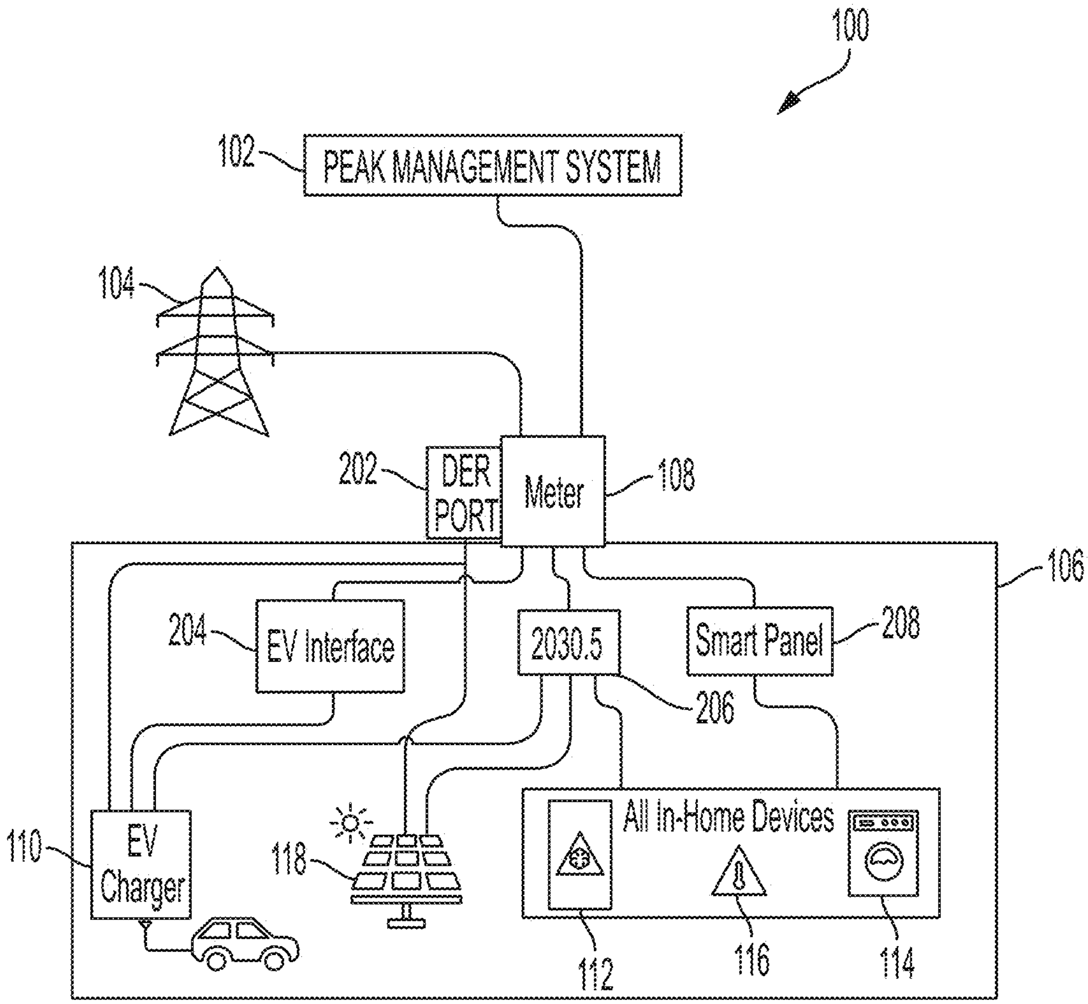
100
102
PEAK MANAGEMENT SYSTEM
104
202
DER PORT
Meter
108
106
204
EV Interface
2030.5
Smart Panel
208
206
110
EV Charger
118
All In-Home Devices
112
116
114

FIG. 2

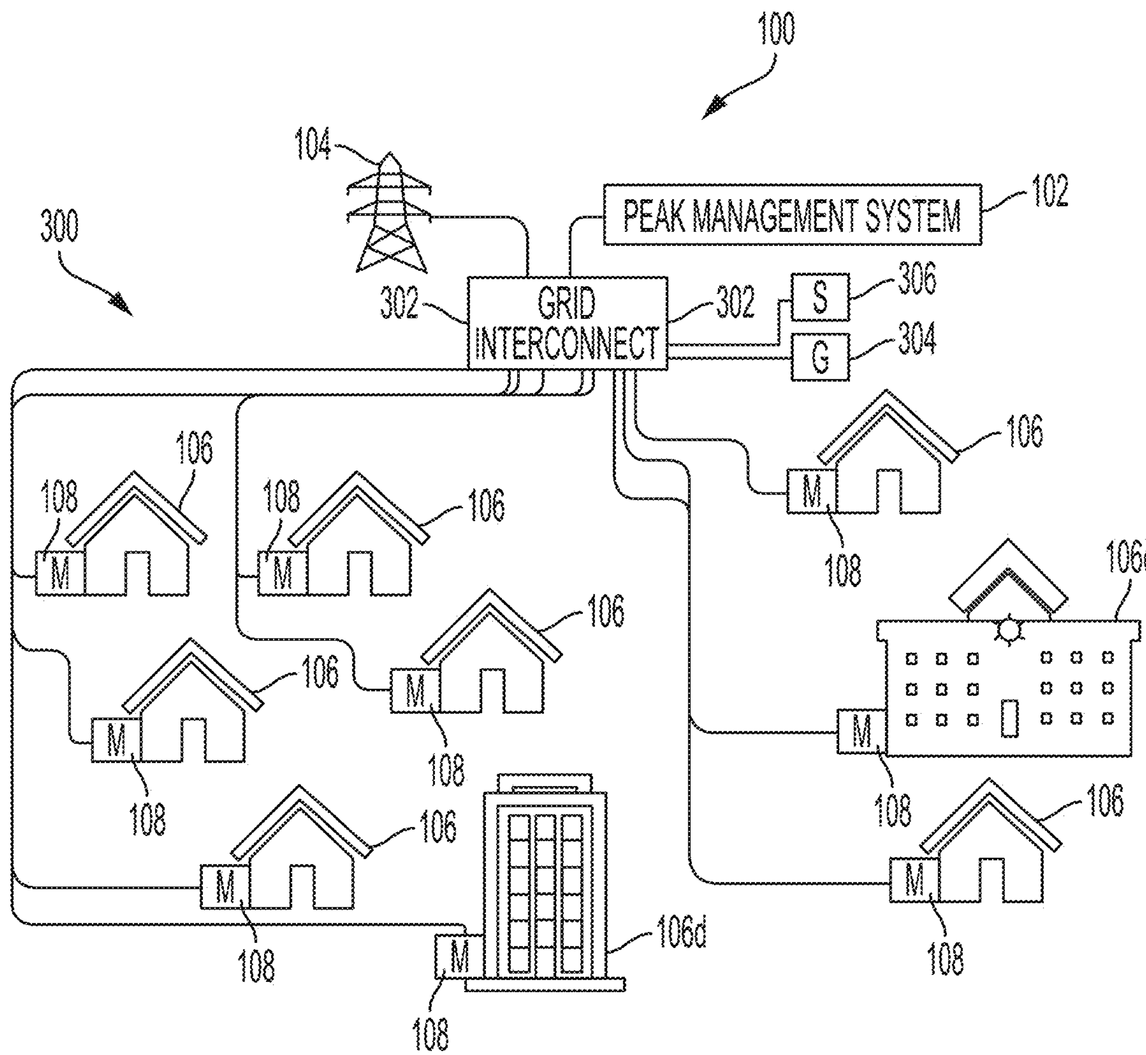
100
104
102
PEAK MANAGEMENT SYSTEM
300
302
GRID INTERCONNECT
302
S
306
G
304
106
108
M
106e
106d

FIG. 3

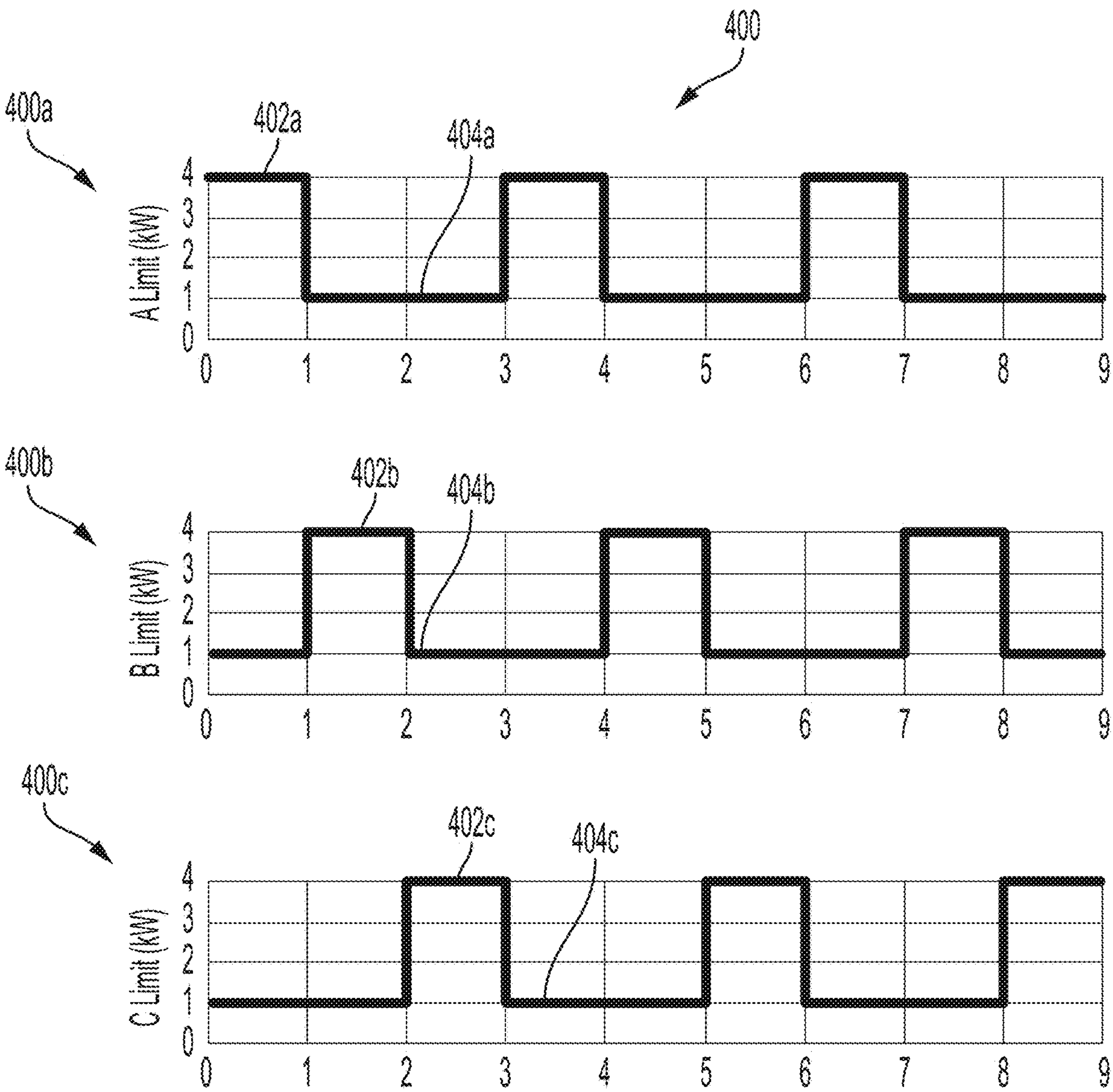
400
400a
402a
404a
A Limit (kW)
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9
400b
402b
404b
B Limit (kW)
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9
400c
402c
404c
C Limit (kW)
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9

FIG. 4

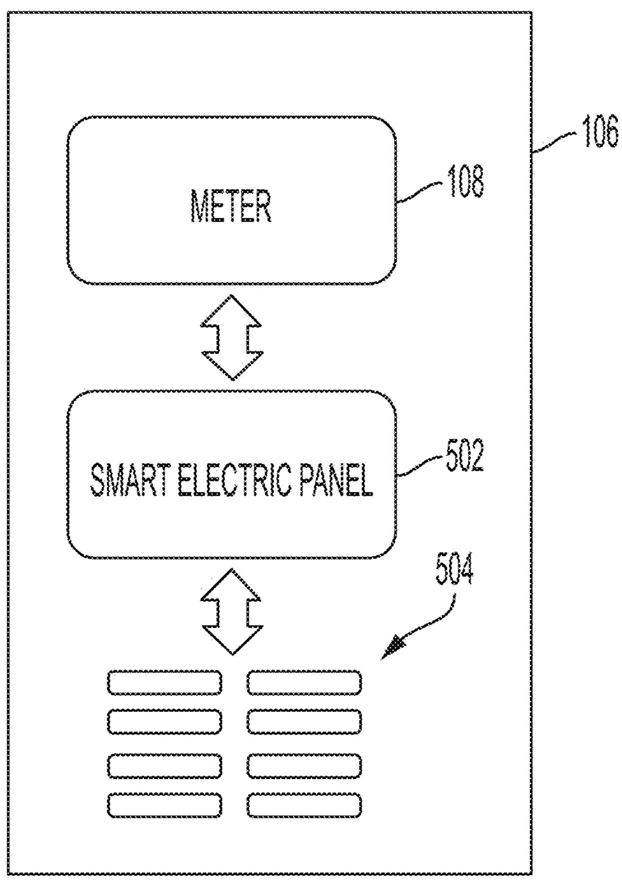

FIG. 5

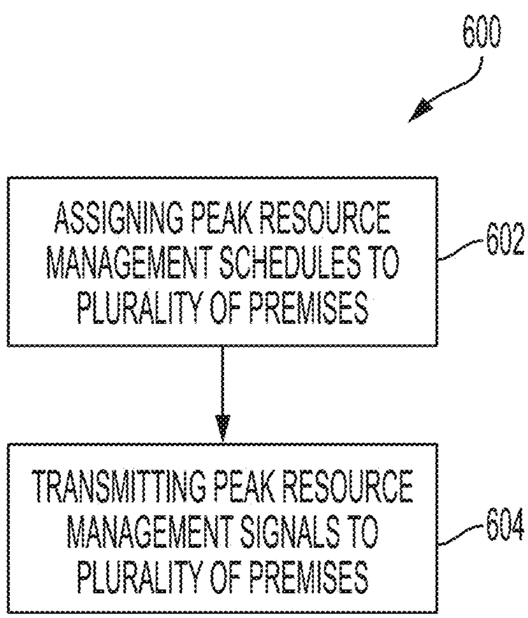
600
ASSIGNING PEAK RESOURCE MANAGEMENT SCHEDULES TO PLURALITY OF PREMISES
602
TRANSMITTING PEAK RESOURCE MANAGEMENT SIGNALS TO PLURALITY OF PREMISES
604

FIG. 6

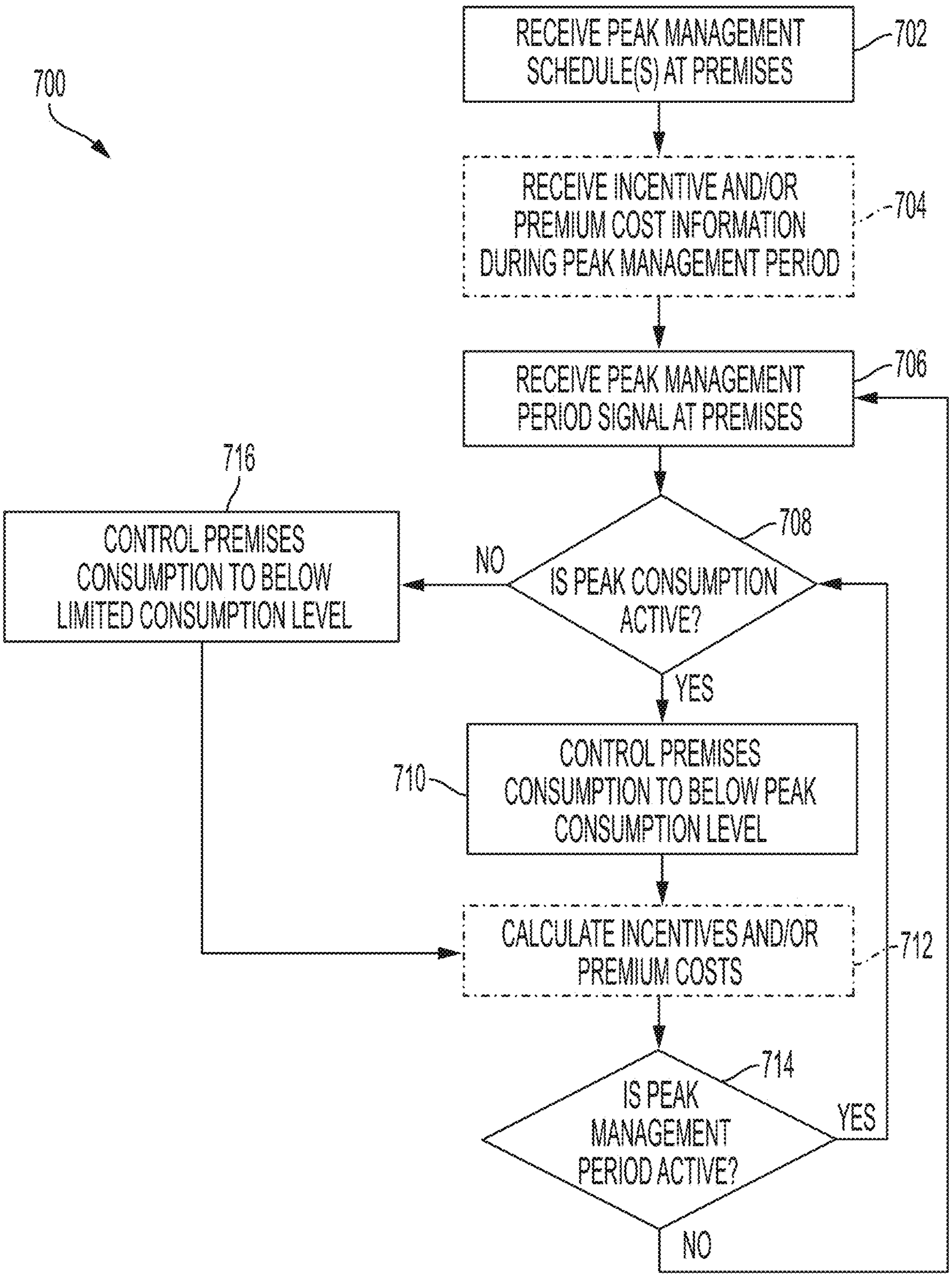
700
RECEIVE PEAK MANAGEMENT SCHEDULE(S) AT PREMISES
702
RECEIVE INCENTIVE AND/OR PREMIUM COST INFORMATION DURING PEAK MANAGEMENT PERIOD
704
RECEIVE PEAK MANAGEMENT PERIOD SIGNAL AT PREMISES
706
716
CONTROL PREMISES CONSUMPTION TO BELOW LIMITED CONSUMPTION LEVEL
NO
IS PEAK CONSUMPTION ACTIVE?
708
YES
710
CONTROL PREMISES CONSUMPTION TO BELOW PEAK CONSUMPTION LEVEL
CALCULATE INCENTIVES AND/OR PREMIUM COSTS
712
714
IS PEAK MANAGEMENT PERIOD ACTIVE?
YES
NO

FIG. 7

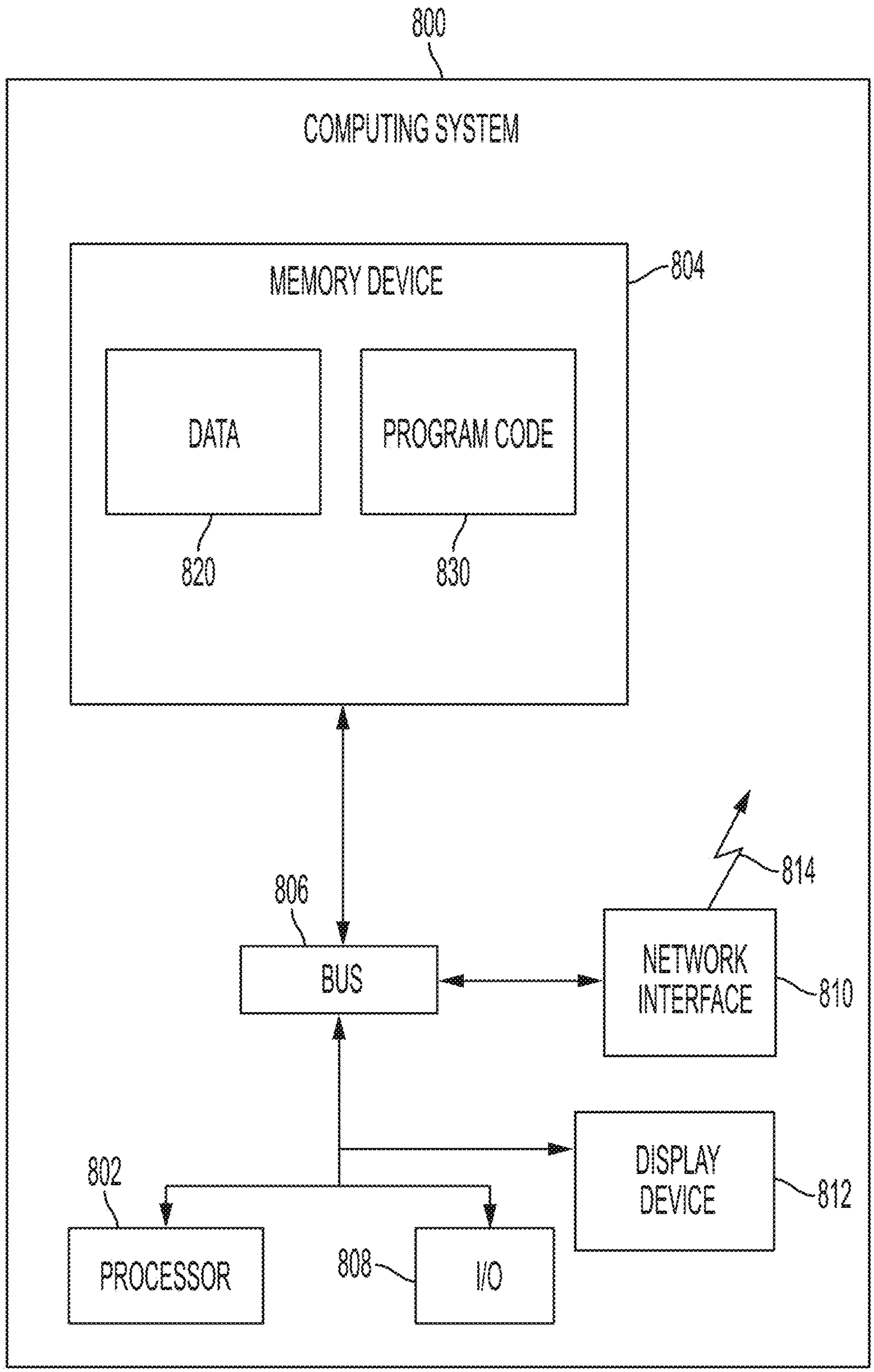
800
COMPUTING SYSTEM
MEMORY DEVICE
804
DATA
PROGRAM CODE
820
830
806
814
BUS
NETWORK INTERFACE
810
DISPLAY DEVICE
812
802
PROCESSOR
808
I/O

FIG. 8

PEAK CONSUMPTION MANAGEMENT FOR RESOURCE DISTRIBUTION SYSTEM

TECHNICAL FIELD

The present disclosure is generally related to consumption management of a resource distribution system. More particularly, but not by way of limitation, the present disclosure is related to managing premises consumption of a resource during a period of peak consumption.

BACKGROUND

In a resource distribution system, such as an electric grid that delivers electric power, extreme weather or critical peak demand periods may result in consumer demand that outpaces resource production capabilities. Because sufficient resources may not be available during these periods to meet the entire resource demand, resource distribution systems may implement rolling outages to avoid system-wide blackouts. Rolling outages may be difficult to manage and generally inequitable across all premises served by the resource distribution system. An alternative to rolling outages to avoid management and inequity concerns may be useful in driving the management of utility assets in a resource distribution system.

SUMMARY

In one implementation, a system includes a metering device, a processor, and a non-transitory, computer-readable memory that includes instructions executable by the processor for causing the processor to perform operations. The operations include accessing a peak resource management schedule. The operations also include receiving a peak resource management signal comprising instructions to commence execution of the peak resource management schedule at a premises. Additionally, the operations include controlling the metering device to monitor premises resource consumption at the premises. Further, the operations include determining that a peak resource consumption portion of the peak resource management schedule is active. The operations further include, in response to determining that the peak resource consumption portion is active, controlling the premises resource consumption at the premises to below a peak resource consumption level of the peak resource management schedule. Furthermore, the operations include determining that a limited resource consumption portion of the peak resource management schedule is active. Moreover, the operations include, in response to determining that the limited resource consumption portion is active, controlling the premises resource consumption at the premises to below a non-zero, limited resource consumption level of the peak resource management schedule.

In another implementation, a computer-implemented method includes assigning, by a peak management system, a peak resource management schedule to a first premises and a second premises within a resource distribution network. The peak resource management schedule includes a first peak consumption period and a first limited resource consumption period for the first premises and a second peak consumption period and a second limited resource consumption period for the second premises, where the first peak consumption period and the second peak consumption period are different. The method also includes transmitting, by the peak management system, a peak resource management signal to the first premises and the second premises.

The peak resource management signal includes instructions to commence execution of the peak resource management schedule at the first premises and the second premises.

In another implementation, a non-transitory computer-readable medium may include instructions that are executable by a processor for causing the processor to perform operations. The operations include accessing a peak resource management schedule. The operations also include receiving a peak resource management signal comprising instructions to commence execution of the peak resource management schedule at a premises. Additionally, the operations include controlling the metering device to monitor premises resource consumption at the premises. Further, the operations include determining that a peak resource consumption portion of the peak resource management schedule is active. The operations further include, in response to determining that the peak resource consumption portion is active, controlling the premises resource consumption at the premises to below a peak resource consumption level of the peak resource management schedule. Furthermore, the operations include determining that a limited resource consumption portion of the peak resource management schedule is active. Moreover, the operations include, in response to determining that the limited resource consumption portion is active, controlling the premises resource consumption at the premises to below a non-zero, limited resource consumption level of the peak resource management schedule.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects, and advantages of the present disclosure are better understood when the following Detailed Description is read with reference to the accompanying drawings.

FIG. **1** illustrates an example of a resource distribution network, according to some implementations described herein.

FIG. **2** illustrates an example of a premises forming part of the resource distribution network of FIG. **1**, according to some implementations described herein.

FIG. **3** illustrates an example of a microgrid forming part of the resource distribution network of FIG. **1**, according to some implementations described herein.

FIG. **4** is an example of a peak resource management schedule assigned to premises of the resource distribution network of FIG. **1**, according to some implementations described herein.

FIG. **5** is a schematic diagram of a metering device controlling premises power consumption with a smart electric panel, according to some implementations described herein.

FIG. **6** is an example flowchart of a process for distributing a peak resource management schedule to premises of a resource distribution network, according to some implementations described herein.

FIG. **7** is an example flowchart of a process for controlling premises power consumption using a peak resource management schedule, according to some implementations described herein.

FIG. **8** is an exemplary computing device used for peak resource management, according to some implementations described herein.

DETAILED DESCRIPTION

The present disclosure describes techniques for peak consumption management of a resource distribution system.

In an example, peak consumption management of a resource distribution system may be provided by distributing a peak resource management schedule from the resource distribution system to premises within the resource distribution system. Each premises within the resource distribution system may be assigned alternating time periods for peak resource consumption and limited resource consumption. Upon receipt of the resource management schedule and instructions to commence execution of the peak resource management schedule at the premises, each premises may control resource consumption in a manner that does not exceed the consumption levels established by the peak resource consumption and the limited resource consumption time periods. By staggering the occurrence of these time periods across all premises within the resource distribution system, a maximum consumption level of the resource distribution system may be reduced from typical resource consumption periods, and the premises may avoid experiencing rolling outages during extreme weather or critical peak consumption periods.

Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

FIG. **1** illustrates an example of a resource distribution network **100**, according to some implementations described herein. The resource distribution network **100** may include a peak management system **102**, a resource distribution system **104**, and a plurality of premises **106***a*, **106***b*, and **106***c*. While the resource distribution network **100** is described herein as a being part of a power distribution environment, other utility systems may incorporate similar peak management system **102**. For example, the peak management system **102** may be employed in a gas, water, or other utility distribution environment.

The resource distribution system **104** may provide power to the premises **106** for consumption by the premises **106**. The amount of power that is distributable to the premises **106** may be limited by a power generation capacity of a power plants that generate power for the resource distribution system **104**. In some examples, extreme weather or critical peak periods may result in consumption demand by the premises **106** that exceeds the power generation capacity of the resource distribution system **104**. To avoid rolling outages at the premises **106** during such periods, the peak management system **102** may implement a peak resource management schedule.

The peak resource management schedule implemented by the peak management system **102** may include schedules of resource consumption by the premises **106** during a peak demand period of the resource distribution network **100**. In an example, the peak resource management schedule may include at least two consumption levels. For example, each of the premises **106** may be assigned a peak resource consumption period and a limited resource consumption period.

In some examples, all of the premises **106** in the resource distribution network **100** may be assigned one of two or more different peak resource management schedules. In such an example, the peak resource consumption periods may be staggered among the multiple peak resource management schedules such that the peak resource consumption periods of the peak resource management schedules do not overlap. In an example with three or more peak resource management schedules, the peak resource consumption periods of the various schedules may not substantially overlap, but the limited resource consumption periods may overlap with other limited resource consumption periods. As used herein, the term "substantially" may refer to a value that is within 10% of another value. Accordingly, if a peak resource consumption period is 1 hour in length, then the active peak resource consumption period may not overlap with other peak resource consumption periods of other peak resource management schedules for more than 6 minutes within any given hour.

The peak management system **102** may assign the peak resource management schedules to metering devices **108***a*, **108***b*, and **108***c* of the premises **106***a*, **106***b*, and **106***c*, respectively. In an example, each of the metering devices **108***a*, **108***b*, and **108***c* may receive a different peak resource management schedule such that the peak resource consumption periods of each of the premises **106***a*, **106***b*, and **106***c* are staggered. When the peak management system **102** activates the peak resource management schedules, the metering devices **108** may control various components of the premises **106** to limit resource consumption at the premises **106** to levels that are below the peak resource consumption level and the limited resource consumption levels depending on an active period of the peak resource management schedule.

For example, the metering device **108***b* may limit operation of devices that contribute the most to resource consumption at the premises when the limited resource consumption period is active. Examples of the devices that may be controlled to a limited amount of consumption may include electric vehicle (EV) chargers **110**, refrigerators **112**, washers and dryers **114**, and heating, ventilation, and air conditioning (HVAC) systems **116**. Other devices may also be limited during the limited resource consumption period. For example, any IoT device that is communicatively coupled to the metering device **108** may be controlled to limit consumption during the limited resource consumption period. In some examples, the metering devices **108** may control devices to not consume power at all when the limited resource consumption period is active. In additional examples, the metering devices **108** may control devices to consume less, but still some, power when the resource consumption period is active. Additionally, the metering device **108** may meter energy producing devices, such as a solar panel **118**, and adjust control of the other premises devices based on a net level of consumption by devices at the premises **106**.

In some examples, an owner, resident, or user of the premises **106** may designate priority levels for various devices at the premises **106** for the metering device **108** to limit resource consumption during the limited resource consumption period. For example, the HVAC system **116** may have a higher priority in a region that experiences high levels of heat when the premises **106** is occupied than the washers and dryers **114**. Accordingly, the metering device **108** may limit resource consumption by the washers and dryers **114** during the limited resource consumption period before controlling HVAC system **116** to reduce resource consumption to a designated level.

During the peak resource consumption period, the metering device **108** may reintroduce devices that were limited during the limited resource consumption period based on priority levels of the devices. For example, the highest priority level device that was previously limited may be fully reactivated before a next highest priority level device. This process may continue until all of the previously limited devices have been reactivated or until a peak resource consumption amount is reached at the premises **106**, whichever occurs first. In other words, during the peak resource consumption period, the premises **106** may consume power at typical resource consumption levels or near typical resource consumption levels.

While the premises **106** in FIG. **1** are each described as being enrolled in the peak resource management schedules, in some examples, not all of the premises **106** of the resource distribution network **100** may be enrolled in the peak resource management schedules. For example, some metering devices within the resource distribution network **100** may not be compatible with the peak resource management schedules or the peak management system **102**. Further, in some examples, the peak resource management schedules among premises **106** within resource distribution groupings of the resource distribution network **100** may be evenly distributed. In other words, by distributing the peak resource management schedules within the distribution groupings, the resource distribution network **100** may ensure that the premises **106** in one resource distribution grouping are not all on the same peak resource management schedule even if a similar number of premises **106** in another resource distribution grouping are on a different peak resource management schedule. This may ensure that various resource distribution groupings of the resource distribution network **100** are not overloaded during implementation of the peak resource management schedules.

FIG. **2** illustrates an example of the premises **106** forming part of the resource distribution network **100**, according to some implementations described herein. To manage the premises load under a limit established by the peak resource management schedule of the premises **106**, the metering device **108** may use various interfaces to control loads at the premises **106** to remain under the peak resource consumption limit and the limited resource consumption limit of the peak resource management schedule. For example, the metering device **108** can include a distributed energy resources (DER) port **202** that provides a mechanism for the metering device **108** to meter resource generation by the solar panel **118**, meter energy flow to or from the EV charger **110**, or meter other distributed energy resource components by the metering device **108**. As discussed above with respect to FIG. **1**, the net consumption detected by the metering device **108** (e.g., premises resource consumption minus premises resource generation) may be used to control consumption by other devices in the premises **106**.

An EV interface **204** may provide communication from the metering device **108** to control the EV charger **110**. Because charging an electric vehicle may consume a significant amount of power, the EV interface **204** may provide a communication interface between the metering device **108** and the EV charger **110** to control times when the EV charger **110** is active. For example, the metering device **108** may control the EV charger **110** to charge an electric vehicle only during peak resource consumption periods when the peak resource management schedule is active. In an example, the metering device **108** can also control, through the EV interface **204**, energy exporting from the electric vehicle through the DER port **202** during a limited resource consumption period. In other words, the electric vehicle may charge and store energy during peak resource consumption periods and sell energy back to the resource distribution system **104** during limited resource consumption periods. In some examples, the peak management system **102** may encourage premises with distributed generation (e.g., solar panels) or storage (e.g., electric vehicles) to export energy at critical times, such as peak times, by offering a price premium on the energy exported from the distributed generation and storage.

In an additional example, an IEEE 2030.5 interface **206** or similar may provide an interface for the metering device **108** to control the loads of the premises **106**. The 2030.5 interface **206** may provide a communication interface for the EV charger **110**, the solar panel **118**, the refrigerator **112**, the washer and dryer **114**, and the thermostat **116**. In an additional example, the metering device **108** may control the refrigerator **112**, the washer and dryer **114**, and the HVAC system **116**, and other large loads at the premises **106** using a smart panel **208**. For example, the metering device **108** can control switches of the smart panel **208** to prevent various loads at the premises **106** from operating, thereby limiting resource consumption at the premises **106** during peak resource consumption periods and limited resource consumption periods. In some examples, the devices of the premises **106** may communicate with the metering device **108** through the interfaces using Wi-Fi, Bluetooth, or other communication protocols. While the EV interface **204**, the 2030.5 interface **206**, and the smart panel **208** are described herein as interfaces for controlling the operation of loads to reduce resource consumption, other interfaces may also be used to perform similar functions.

In some examples, implementation of the peak resource management schedule by the metering device **108** may take into account expected demand for controlled loads of the premises **106** (e.g., the EV charger **110**, the refrigerator **112**, the washer and dryer **114**, and the HVAC system **116**) and uncontrolled loads of the premises **106** (e.g., any other loads that are not controllable directly by the metering device **108**). Further, the peak management system **102** may provide incentives that encourage energy exporting from the solar panels **118** or onsite energy storage devices including battery storage, electric vehicles, and other energy storage devices during critical peak periods. For example, the peak management system **102** may provide an enhanced rate for energy exporting during critical peak periods to encourage an owner of a premises **106** to sell energy back to the resource distribution system **104**. Additionally, the peak management system **102** may provide a price structure that incentivizes a reduction in resource consumption below specified limits by providing monetary rewards for specific actions, such as maintaining resource consumption less than a further limited level from the peak resource management schedule.

Further, in some examples, the peak resource management schedule may include limiting specific loads during the peak resource consumption period. For example, the peak resource management schedule may include an indication that the EV charger **110** cannot draw power or cannot draw full power from the resource distribution network **104** during the peak resource consumption period. Other devices may also be similarly limited during the peak resource consumption period.

FIG. **3** illustrates an example of a microgrid **300** forming part of the resource distribution network **100**, according to some implementations described herein. The microgrid **300** may include the peak management system **102**, the resource distribution system **104**, and a plurality of premises **106**. The resource distribution system **104** may provide power to the premises **106** for consumption by the premises **106**. The premises **106** may include homes, hospitals, retail stores, grocery stores, restaurants, office buildings, schools, municipal buildings (e.g., police stations, fire stations, courthouses, etc.), or any other buildings making up the microgrid **300** with metering devices **108**. The amount of power that is distributable to the premises **106** from the resource distribution system **104** may be limited by a power generation capacity of a power plant that generate power for the resource distribution system **104**. In some examples, extreme weather or critical peak periods may result in consumption demand by the premises **106** that exceeds the power generation capacity of the resource distribution system **104**. To avoid rolling outages at the premises **106** during such periods, the peak management system **102** may island the microgrid **300** and implement a peak resource management schedule. For example, the peak management system **102** may control a grid interconnect **302** between the microgrid **300** and the resource distribution system **104**. The grid interconnect **302** enables isolation between the microgrid **300** and the resource distribution system **104** during critical resource periods. In some examples, the microgrid **300** may receive a microgrid resource management schedule from the grid-wide peak management system that assigns consumption limits for the microgrid **300** and other premises **106** that are not part of the microgrid **300**. In some examples, the microgrid resource management schedule received from the grid-wide peak management system may establish a zero-consumption level for the microgrid **300** from the resource distribution system **104** for certain time periods. In other words, during the zero-consumption level, any power consumed at the microgrid **300** is generated or otherwise accessed from distributed energy resources of the microgrid **300**. Additionally, the peak management system **102** of the microgrid **300** may provide peak resource management schedules to each premises **106** of the microgrid **300** based on the microgrid resource management schedule.

The microgrid **300** may have sufficient power generation **304** and energy storage **306** to maintain the premises **106** in the microgrid **300** at a limited overall load. The power generation **304** may include any type of distributed energy generation by the microgrid **300**, such as solar, wind, and hydro power plants, as well as fuel-based generation (e.g., natural gas and diesel generation) and other fuel and renewable generation sources. The energy storage **306** can be any energy storage devices associated with the microgrid **300**. In some examples, the energy storage **306** may include electric vehicles at the premises **106** or other battery cells capable of storing energy. In some examples, one or more of the premises **106** in the microgrid **300** may include the power generation **304** and the energy storage **306**. Management of the power generation **304** and the energy storage **306** at the premises **106** may be performed at the individual premises level and as an aggregated resource of the microgrid **300**.

The peak management system **102** may further limit an overall load of the microgrid **300** by implementing a peak resource management schedule for the microgrid **300**. Similar to the peak resource management schedule discussed above with respect to FIG. **1**, the peak resource management schedule implemented by the peak management system **102** may include schedules of resource consumption by the premises **106** during a peak demand period of the resource distribution network **100**. The peak resource management schedule may include at least two consumption levels. For example, each of the premises **106** may be assigned a peak resource consumption period and a limited resource consumption period.

In some examples, all premises **106** in the resource distribution network **100** may be assigned one of two or more different peak resource management schedules. In such an example, the peak resource consumption periods may be staggered among the multiple peak resource management schedules such that the peak resource consumption periods of the peak resource management schedules do not overlap. In an example with three or more peak resource management schedules, the peak resource consumption periods of the various schedules may not substantially overlap, but the limited resource consumption periods may overlap with other limited resource consumption periods. As used herein, the term "substantially" may refer to a value that is within 10% of another value. Accordingly, if a peak resource consumption period is 1 hour in length, then the active peak resource consumption period may not overlap with other peak resource consumption periods of other peak resource management schedules for more than 6 minutes within any given hour.

The peak management system **102** may assign the peak resource management schedules to metering devices **108** of the premises **106**. In an example, each of the metering devices **108** may receive one of several different peak resource management schedules such that the peak resource consumption periods are divided among the premises **106**. When the peak management system **102** activates the peak resource management schedules, the metering devices **108** may control various components of the premises **106** to limit resource consumption at the premises **106** to levels that are below the peak resource consumption level and the limited resource consumption levels depending on an active period of the peak resource management schedule. The result may be that the overall consumption of the microgrid **300** remains below a level that is supportable by the power generation **304** of the microgrid **300** when the microgrid is islanded from the resource distribution system **104**. In some examples, the microgrid **300** may export unused power generated by the power generation **304** or stored at the energy storage **306** to the resource distribution system **104** in exchange for a premium price of the energy during a critical resource period. In an additional example, the microgrid **300** may be provided with a schedule from the grid-wide peak management system that establishes certain time periods where the microgrid **300** is expected to export energy back to the grid. The schedule may be established by monetary incentives to the microgrid **300** or by an agreement for the microgrid **300** to provide a certain amount of energy per unit of time.

The premises level control of the microgrid **300** may be controlled in a manner similar to the premises level control of the premises **106** that are not in a microgrid, as discussed above with respect to FIGS. **1** and **2**. For example, the metering devices **108** may limit operation of devices that contribute the most to resource consumption at the premises **106** when the limited resource consumption period is active. Examples of the devices that may be controlled to a limited amount of consumption may include electric vehicle chargers, refrigerators, washers and dryers, and heating, ventilation, and air conditioning (HVAC) systems. Other devices may also be limited during the limited resource consumption period. For example, any IoT device that is communicatively coupled to the metering device **108** may be controlled to limit consumption during the limited resource consumption period. Additionally, the metering device **108** may meter energy producing devices, such as a solar panel, and adjust control of the other premises devices based on a net level of consumption by devices at the premises **106**.

In some examples, an owner, resident, or user of the premises **106** may designate priority levels for various devices at the premises **106** for the metering device **108** to limit resource consumption during the limited resource consumption period. For example, the HVAC system **116** may have a higher priority in a region that experiences high levels of heat when the premises **106** is occupied than the washers and dryers **114**. Accordingly, the metering device **108** may limit resource consumption by the washers and dryers **114** during the limited resource consumption period before controlling HVAC system **116** to reduce resource consumption to a designated level.

During the peak resource consumption period, the metering device **108** may reintroduce devices that were limited during the limited resource consumption period based on priority levels of the devices. For example, the highest priority level device that was previously limited may be fully reactivated before a next highest priority level device. This process may continue until all of the previously limited devices have been reactivated or until a peak resource consumption amount is reached at the premises **106**, whichever occurs first. In other words, during the peak resource consumption period, the premises **106** may consume power at typical resource consumption levels or near typical resource consumption levels.

In some examples, some of the premises **106** in the microgrid **300**, or in the overall resource distribution network **100**, may not be subject to the peak resource management schedules. For example, a premises **106***d*, which may be a commercial building, may not be on the peak resource management schedules, or may have a different triggering event for placement on the peak resource management schedules. Similarly, a premises **106***e*, which may be a school or a hospital, may not be on the peak resource management schedules. By avoiding the peak resource management schedules for select premises **106***d* and **106***e*, some premises that provide critical infrastructure may continue to operate under normal operating conditions to meet a critical infrastructure goal of the premises.

FIG. **4** is an example of a peak resource management schedule **400** assigned to the premises **106** of the resource distribution network **100**, according to some implementations described herein. As depicted, the peak resource management schedule **400** include three individual schedules **400***a*, **400***b*, and **400***c*. Each of the schedules **400** may be assigned to a number of premises **106** within the resource distribution network **100** or within the microgrid **300**. While three individual schedules are depicted in the peak resource management schedule **400**, more or fewer individual schedules can also be used for the premises **106** of the resource distribution network **100**. At a microgrid level, similar schedules to the schedules **400** may be applied. In some examples, amounts of permissible energy consumption may be established for the premises **106** of the microgrid **300** that are different from amounts of permissible energy consumption for the premises **106** of the resource distribution network **100**.

Each of the individual schedules **400***a*, **400***b*, and **400***c* include peak resource consumption periods **402***a*, **402***b*, and **402***c*, respectively, and limited resource consumption periods **404***a*, **404***b*, and **404***c*, respectively. The peak resource consumption periods **402** and the limited resource consumption periods **404** may be defined by a power limit in kilowatts over a specified period of time. As illustrated in the example depicted in FIG. **4**, the peak resource consumption periods **402** have a limit of four kilowatts, and the limited resource consumption periods **404** have a limit of one kilowatt. When the individual schedules **400***a*, **400***b*, and **400***c* are evenly divided among all of the premises **106** within the resource distribution network **100**, the average consumption limit per premises **106** of the resource distribution network **100** is two kilowatts. This average consumption limit may be increased or decreased by adjusting the limits of the peak resource consumption periods **402** and the limited resource consumption periods **404**, the timing of the peak resource consumption periods **402** and the limited resource consumption periods **404**, or a combination thereof. While FIG. **4** depicts the schedules **400** with specific limit values and times, the specific values are for illustration purposes only. Other specific limit values and times may be used within the schedules **400** depending on parameters of the resource distribution network **100**.

In an example, each of the premises **106** in the resource distribution network **100** may be assigned one of the individual schedules **400***a*, **400***b*, and **400***c* upon installation of a metering device **108** at a premises **106**. In an additional example, the peak management system **102** may assign the individual schedules **400***a*, **400***b*, and **400***c* to the premises **106** upon initiating the peak resource management schedule **400**. In some examples, the premises **106** with storage capabilities, such as electric vehicles, stationary batteries, and other energy storage devices or systems, may utilize the peak resource consumption periods **402** to charge batteries and discharge the batteries during the limited resource consumption periods **404** to maintain a more consistent level of resource consumption during the totality of the peak resource management schedule **400**. In some examples, the schedules **400** may be part of an opt-in system, where the premises **106** are able to opt into receiving one of the individual schedules **400***a*, **400***b*, and **400***c* in exchange for incentives from a utility.

In some examples, multiple, different peak resource management schedules **400** may be assigned to the premises **106**. For example, the different schedules **400** may be assigned based on an expected severity of the critical resource period. For example, the peak resource management system **102** may initiate a particular schedule **400** when a high temperature condition for a particular day is expected to be extreme, and the peak resource management system **102** may initiate a different particular schedule **400** when a high temperature condition for a different particular day is expected to be less than the extreme temperature condition. In such an example, the extreme weather schedule **400** may have a shorter or lower peak resource consumption period **402** and a longer or lower limited resource consumption period **404** than the peak resource consumption period **402** and limited resource consumption period **404** of the schedule **400** initiated on a more moderate weather day.

FIG. **5** is a schematic diagram of a metering device **108** controlling premises power consumption with a smart electric panel **502**, according to some implementations described herein. The metering device **108** may include a peak resource management schedule **400**. To maintain the premises **106** within the limits of the peak resource management schedule **400**, the metering device **108** may control the smart electric panel **502** to enable provision of power to various components **504** of the premises **106** controlled by the smart electric panel **502**. For example, the components **504** can include individual components of the premises **106** such as an HVAC system, refrigerators, washers and dryers, etc. that may be controllable by an individual circuit breaker of the smart electric panel **502**. In additional examples, the components **504** may include rooms or other-sized regions of the premises **106** that are controllable by individual circuit breakers of the smart electric panel **502**.

The metering device **108** may control the smart electric panel **502** to remove application of power from particular components **504** at the premises **106** to maintain the premises **106** below the consumption limits of the peak resource management schedule **400**. In some examples the smart electric panel **502** may prioritize the components **504** and intelligently control the application or removal of power to the various components **504** based on the priority of the components **504**. In some examples, the metering device **108** can control the smart electric panel **502** while also directly controlling individual components of the premises **106**. For example, the metering device **108** can control one or more resource consuming devices to reduce resource consumption to zero or to a non-zero level. The non-zero level may include a reduction from a peak consumption level of the one or more resource consuming devices.

FIG. **6** is an example flowchart of a process **600** for distributing the peak resource management schedule **400** to premises of the resource distribution network **100**, according to some implementations described herein. At block **602**, the process **600** involves assigning peak resource management schedules **400** to a plurality of premises **106**. The plurality of premises **106** may include all or substantially all of the premises **106** within the resource distribution network **100**. In an example, each of the premises **106** may be assigned one of a plurality of individual schedules that make up the peak resource management schedule **400**. The premises **106** may be evenly or substantially evenly divided among the individual schedules of peak resource management schedule **400** to maintain equity among all of the premises **106** within the resource distribution network **100** while maintaining overall consumption of the resource distribution network **100** below resource producing capabilities of the resource distribution network **100**.

At block **604**, the process **600** involves transmitting peak resource management signals to the plurality of premises **106** of the resource distribution network **100** to initiate the peak resource management schedules **400** at the premises **106**. For example, the peak resource management system **102** may transmit the peak resource management signals to the metering devices **108** of the premises **106**. In some examples, the peak resource management system **102** may transmit the signals through a mesh network of the metering devices within the resource distribution network **100**. In additional examples, other communication types, such as cellular communication networks, wide area networks, or combinations of communication networks, may be used to transmit the signals to the metering devices **108**.

The peak resource management signals may be transmitted by the peak resource management system **102** when a determination is made that a critical resource period has commenced or will commence imminently. The signal may be an indication to the metering devices **108** that the peak resource management schedules **400** should commence. In some examples, the signal may be a single broadcast command from the peak resource management system **102** to initial the schedules **400** at all metering devices **108** within the resource distribution network **100**. In an example, the broadcast time for a mesh network of the metering devices **108** may take approximately 10 minutes to reach all endpoints. Each of the metering devices **108** may be assigned a particular peak resource management schedule **400**. The schedules **400** may be assigned based on local area network identification (LAN ID), geographical locations, or other parameters of the metering devices **108**. The broadcast message from the peak resource management system **102** may be a command to initiate the pre-assigned schedules **400** at the metering devices **108**.

FIG. **7** is an example flowchart of a process **700** for controlling premises power consumption using the peak resource management schedule **400**, according to some implementations described herein. At block **702**, the process **700** involves receiving one or more peak resource management schedules **400** at a premises **106**. In an example, a metering device **108** of the premises **106** may be preassigned with the one or more peak resource management schedules **400**. In additional examples, the metering device **108** may receive the schedules **400** from the peak management system **102**.

At block **704**, the process **700** involves receiving incentive information premium cost information, or both during a peak resource management period. The incentive information may include premium prices for selling energy back to the resource distribution network **100** during a peak management period, and the premium cost may include premium costs for exceeding usage limits established by the peak resource management schedules **400** during the peak resource management period. In some examples, the metering device **108** of the premises **106** may intelligently control consumption of by the premises **106** taking into account the premium information and the cost information.

At block **706**, the process **700** involves receiving a peak management period signal at the premises **106**. The peak management period signal may be an instruction for the metering device **108** to initiate the peak resource management schedule **400**. In some examples, the peak management period signal may include an indication of a particular point in time in which the peak resource management schedule **400** should commence. By identifying the particular point in time for commencing the peak resource management schedule **400**, each of the premises **106** may commence the schedule **400** in a manner that avoids unintended overlapping of peak resource consumption periods **402** for the premises **106** that include different schedules **400**. In an example where the metering device **108** includes multiple schedules **400**, the peak management period signal may identify a particular schedule **400** for the metering device **108** to initiate.

At block **708**, the process **700** involves determining whether a peak resource consumption period **402** of the peak resource management schedule **400** is active at the premises **106**. If the peak resource consumption period is active, then, at block **710**, the process **700** involves controlling consumption of energy at the premises **106** below a peak consumption limit established by the peak resource consumption period **402** of the peak resource management schedule **400**. In an example, the metering device **108** may take into account priority designations of various loads in the premises **106** to meet the peak consumption limit. For example, a higher priority load may remain in a consuming state until all lower priority loads have already been removed to meet the peak consumption limit.

At block **712**, the process **700** involves calculating incentives for exporting power to the resource distribution network **100**, premium costs for exceeding any resource consumption limits established by the peak resource management schedule **400**, or both. For example, when the premises **106** sells power back to the resource distribution network **100**, the incentive may be calculated based on the premium price associated with the peak management period. Likewise, when the premises **106** exceeds consumption limits of the schedule **400**, the premium cost may be calculated based on a premium associated with exceeding the consumption limits. In some examples, this information collected at block **712** may be transmitted to a headend system along with other metrology information for generation of bills associated with resource consumption at the premises **106**. In additional examples, this information may be used to adjust control of the consumption at the premises **106** during future peak management periods.

At block **714**, the process **700** involves determining if the peak management period is still active. In some examples, the peak management period may continue for a preset period of time. In additional examples, the peak management period may stop with the peak management system **102** transmits an additional signal to cancel the peak management period (e.g., when a critical resource period is determined to be over). If the peak management period is still active, then the process **700** may return to block **708** to determine if the peak resource consumption period **402** is active. If the peak management period is no longer active, then the process **700** may return to block **706** to await an additional peak management period signal to initiate the peak resource management schedule **400** at the premises **106**. When the process **700** returns to block **706** to await the additional peak management period signal, the process **700** may further include controlling the premises **106** to resume a standard consumption level.

If, at block **708**, a determination is made that the peak resource consumption period **402** is not active, then, at block **716**, the process **700** involves controlling the energy consumption at the premises **106** to below a limited consumption level limit established by the limited resource consumption period **404** of the peak resource management schedule **400**. The process **700** may then return to block **712** to calculate incentives and costs associated with resource generation or consumption at the premises **106**.

FIG. **8** illustrates an exemplary computing device used for managing premises consumption of a resource during a period of peak consumption, according to some implementations described herein. Any suitable computing system may be used for performing the operations described herein. The depicted example of a computing device **800** includes a processor **802** communicatively coupled to one or more memory devices **804**. The processor **802** executes computer-executable program code **830** stored in a memory device **804**, accesses data **820** stored in the memory device **804**, or both. Examples of the processor **802** include a microprocessor, an application-specific integrated circuit ("ASIC"), a field-programmable gate array ("FPGA"), or any other suitable processing device. The processor **802** can include any number of processing devices or cores, including a single processing device. The functionality of the computing device may be implemented in hardware, software, firmware, or a combination thereof.

The memory device **804** includes any suitable non-transitory computer-readable medium for storing data, program code, or both. A computer-readable medium can include any electronic, optical, magnetic, or other storage device capable of providing a processor with computer-readable instructions or other program code. Non-limiting examples of a computer-readable medium include a flash memory, a ROM, a RAM, an ASIC, or any other medium from which a processing device can read instructions. The instructions may include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, including, for example, C, C++, C#, Visual Basic, Java, or scripting language.

The computing device **800** may also include a number of external or internal devices, such as input or output devices. For example, the computing device **800** is shown with one or more input/output ("I/O") interfaces **808**. An I/O interface **808** can receive input from input devices or provide output to output devices. One or more busses **806** are also included in the computing device **800**. The bus **806** communicatively couples one or more components of a respective one of the computing device **800**.

The computing device **800** executes program code **830** that configures the processor **802** to perform one or more of the operations described herein. For example, the program code **830** causes the processor to perform the operations described in FIGS. **1-7**.

The computing device **800** also includes a network interface device **810**. The network interface device **810** includes any device or group of devices suitable for establishing a wired or wireless data connection to one or more data networks. The network interface device **810** may be a wireless device and have an antenna **814**. The computing device **800** can communicate with one or more other computing devices implementing the computing device or other functionality via a data network using the network interface device **810**. In some examples, the interface device **810** may provide cellular network communication functionality, mesh network communication functionality, or any other communication functionalities for the computing device **800**.

The computing device **800** can also include a display device **812**. Display device **812** can be a LCD, LED, touchscreen or other device operable to display information about the computing device **800**. For example, information could include an operational status of the computing device, network status, etc.

In an example, the computing device **800** may form part of a metering device, such as the metering device **108**. In such an example, the data **820** stored in the memory device **804** may include metrology data. Further, the program code **830** may include program code that is accessible by the processor **802** to perform metrology operations associated with the metering device **108**.

While the present subject matter is described in detail with respect to specific aspects thereof, it will be appreciated that those skilled, upon attaining an understand of the foregoing and the following, may readily produce alternations to, variations of, and equivalents to such aspects. Accordingly, it should be understood that the present disclosure has been presented for purposes of example rather than limitation and does not preclude inclusion of such modifications, variations, and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art.

What is claimed is:

1. A system comprising:

a metering device, including a distributed energy resource (DER) port, configured to meter resource generation by one or more DER components;

a smart panel configured to control application of power to a plurality of devices;

a processor; and

a non-transitory, computer-readable memory that includes instructions executable by the processor for causing the processor to:

access a peak resource management schedule;

receive a peak resource management signal comprising instructions to commence execution of the peak resource management schedule at a premises;

control the metering device to monitor premises resource consumption at the premises;

determine that a peak resource consumption portion of the peak resource management schedule is active;

in response to determining that the peak resource consumption portion is active, control the premises resource consumption at the premises to an amount that is less than a peak resource consumption level of the peak resource management schedule, the controlling the premises resource consumption at the premises comprising controlling the application of power by the smart panel to the plurality of devices;

determine that a limited resource consumption portion of the peak resource management schedule is active; and

in response to determining that the limited resource consumption portion is active, control the premises resource consumption at the premises to an amount that is less than a non-zero, limited resource consumption level of the peak resource management schedule.

2. The system of claim **1**, where the peak resource management schedule comprises:

a first peak consumption period and a first limited resource consumption period for a first premises associated with the metering device, where a second peak consumption period and a second limited resource consumption period for a second premises associated with a second metering device are different from the first peak consumption period and the second peak consumption period.

3. The system of claim **2**, where the first peak consumption period and the second peak consumption period do not overlap by more than 10% of either the first or second peak consumption period.

4. The system of claim **1**, where the instructions are further executable by the processor for causing the processor to:

receive incentive information associated with controlling the premises resource consumption to an amount that is less than a specified level;

receive premium cost information associated with controlling the premises resource consumption to above the specified level; and

control the premises resource consumption using the incentive information and the premium cost information.

5. The system of claim **1**, where the peak resource consumption portion enables the premises resource consumption to be an amount that is twice the amount of premises resource consumption during the limited resource consumption portion, or more than twice the amount of premises resource consumption during the limited resource consumption portion.

6. The system of claim **1**, where controlling the premises resource consumption to an amount that is less than the non-zero, limited resource consumption level of the peak resource management schedule comprises:

assigning priority levels to a plurality of resource consuming devices; and

instructing a subset of the plurality of resource consuming devices with a lowest priority level of the priority levels to reduce resource consumption.

7. The system of claim **1**, where the premises comprises a portion of a microgrid system, a first controller of the microgrid system receives a microgrid resource management schedule from a second controller of a resource distribution network, and the first controller of the microgrid system is configured to provide the peak resource management schedule to each premises of the microgrid system based on the microgrid resource management schedule.

8. A non-transitory, computer-readable medium comprising instructions that are executable by a processor for causing the processor to perform operations comprising:

accessing a peak resource management schedule;

receiving a peak resource management signal comprising instructions to commence execution of the peak resource management schedule at a premises;

controlling a metering device to monitor premises resource consumption at the premises;

determining that a peak resource consumption portion of the peak resource management schedule is active;

in response to determining that the peak resource consumption portion is active, controlling the premises resource consumption at the premises to an amount that is less than a peak resource consumption level of the peak resource management schedule, where controlling the premises resource consumption at the premises comprises controlling an application of power by a smart panel to a plurality of devices;

determining that a limited resource consumption portion of the peak resource management schedule is active; and

in response to determining that the limited resource consumption portion is active, controlling the premises resource consumption at the premises to an amount that is less than a non-zero, limited resource consumption level of the peak resource management schedule.

9. The non-transitory, computer-readable medium of claim **8**, where the peak resource management schedule comprises:

a first peak consumption period and a first limited resource consumption period for a first premises associated with the metering device, where a second peak consumption period and a second limited resource consumption period for a second premises associated with a second metering device are different from the first peak consumption period and the second peak consumption period.

10. The non-transitory, computer-readable medium of claim **9**, where the first peak consumption period and the second peak consumption period do not overlap by more than 10% of either the first or second peak consumption period.

11. The non-transitory, computer-readable medium of claim **8**, the operations further comprising:

receiving a second peak resource management signal comprising instructions to cancel the peak resource management schedule at the premises; and

in response to receiving the second peak resource management signal, controlling the premises to resume a standard resource consumption level.

12. The non-transitory, computer-readable medium of claim **8**, where the operation of controlling the premises resource consumption to the amount that is less than the non-zero, limited resource consumption level of the peak resource management schedule comprises:

instructing one or more resource consuming devices to reduce resource consumption to a zero level or to a non-zero level, where the non-zero level comprises a reduction from a peak consumption level of the one or more resource consuming devices.

13. The non-transitory, computer-readable medium of claim **8**, where the instructions to commence the execution of the peak resource management schedule at the premises further comprise a particular time in which each premises that is assigned the peak resource management schedule is to commence the execution of the peak resource management schedule.

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